CIRC HQ STUDENT HANDOUT | GRADE 5

KidWind 3: Build the best set

Goal: I can use class evidence to build a better set of blades.

35 minutes | 7 steps | Engineering and Design | Playbook A meeting 12, grade 5 path; Playbook B meeting 13, grade 5 path

Ask | Plan | Build | Test | Improve

You are done when

Compare your best set with your first set, in centimeters.

You will use

- One teacher-run test station, built and piloted first from parts on hand
- One guarded fan at one fixed setting, with tape marks and a ruler
- A smooth, blunt shaft turning in a larger straw or tube, with a cork or foam hub
- String on a fixed spool, one small cup and the same load pieces every trial
- Cardstock or index cards, reusable blade supports, tape and student scissors
- A marked blade-angle example, timer, shallow catch tray, team folders and record sheets
- Printed practice data table for the no-fan paper path

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 2 min

- Sit with your team.
- Read the goal: use evidence to build better.

2 Step 2: Find your records — 1 min

- Find your Set A and Set B results.

3 Step 3: Read the class chart — 5 min

- Look for the sets that lifted the cup highest.
- What did those blades have in common?

4 Step 4: Build your best set — 12 min

- Use the evidence to design one best set.
- Say which chart result supports your choice.

5 Step 5: Repeat fair trials — 10 min

- Run two or three 10-second trials, as your teacher directs.
- Record every result in centimeters, even the low ones.

6 Step 6: Reset the station — 3 min

- Fan off first.
- Rotor sets go flat in your team folder.
- Cup and load pieces go back in the station tray.

7 Step 7: Compare — 2 min

- Compare your best set with your first set, in centimeters.

What to turn in

Your best set's results from two or three 10-second trials, as your teacher directs, in centimeters and compared with your first set.

No computer? Another way to do it

No-fan paper path: follow the paper steps and the labeled practice data below. It uses the same measurement but does not claim you built or tested a windmill.

Finished early? Next challenge

Test one blade angle change on your best set and record it separately.

Stay safe

Only the teacher runs the fan and prepares any holes with pointed tools. Stay out of the rotor's spin path. Fan off before anyone touches blades, the cup or the string. Blades are paper or cardstock only. Shaft ends are blunt.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.

No-fan paper path

Use these paper steps instead when your teacher chooses the no-fan path.

Goal: I can explain how wind can lift a load and use sample results to plan a fair test.

1. Compare design A with design C.
2. Choose a design to test next.
3. Use two table numbers to explain your choice.
4. Draw the next test.

Keep: A prediction backed by the table. It is not a measured new result.

Practice data created for this lesson. These are not student measurements or evidence about a real blade design.

Unit: Centimeters lifted in 10 seconds with one unchanged cup and load

Practice data created for this lesson. These are not student measurements or evidence about a real blade design.

Practice design	Only stated change	Trial 1	Trial 2	Trial 3
A	Baseline	5	10	5
B	Blade angle only	10	15	10
C	Blade size only, compared with A	10	10	15